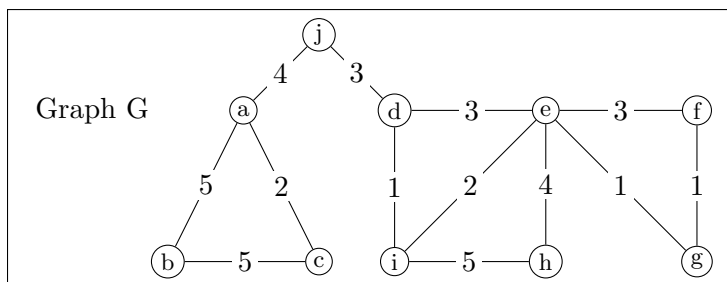


Math F113X: Homework Set 5

Answer the following problems from the Graph Theory section in the order assigned.

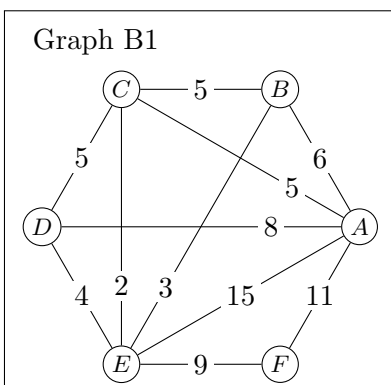
A, B, 2, 23, C, D

Problem A: Use the drawing of Graph G (in box) to answer the questions.

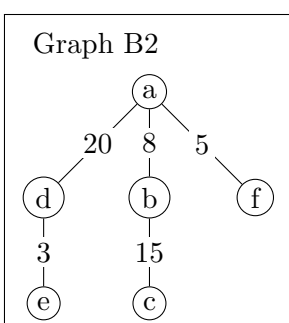


1. Draw two different spanning trees of Graph G.
2. Determine the total weight of each tree from part 1 above.
3. Find a minimum weight spanning tree. (Use any method you like, including guess-and-check.)

Problem B: Use Kruskal's Algorithm to find a minimum cost spanning tree in each graph below. Make sure to show appropriate work. A good way to do this is to make a table like the one below. List every edge in ascending order of weight (that is, smallest to largest) in the first column. List the corresponding weight in the second column. In the third column, use a checkmark to indicate that you are using the edge in your spanning tree and an "x" to indicate that you are not.



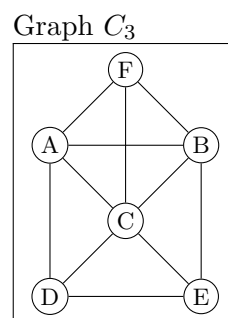
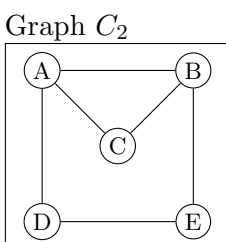
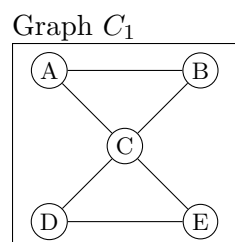
edge	weight	used?



edge	weight	used?

Problem C: Describe one real-world situation in which it would be useful to find a minimal spanning tree for a graph. Be sure to clearly describe the overall scenario, specify what the vertices represent, specify what the edges represent, specify what the edge weights represent, and state why you would want to find a minimal weight spanning tree.

Problem D: For each graph below, (i) determine if it has an Euler circuit and (ii) determine if it has an Euler path. Justify your answer.



Remember to write up your homework solutions according to the homework writeup guidelines.

Homework is graded using the following rubric for each problem (or problem part):

2 points: You provided a complete answer, with supporting work, written up clearly

1 point: Some attempt at a solution, but incomplete writeup / unclear / illegible / no answer

1 point: Only an answer, with no supporting work

0 points: Missing.

After you do the homework, you need to check your answers against the solutions! Then figure out your errors (if any) and revise your homework before you submit it.